



BITS Pilani
Pilani Campus

CE F230 **Civil Engineering Materials**

Concrete: **Cement and hydration**

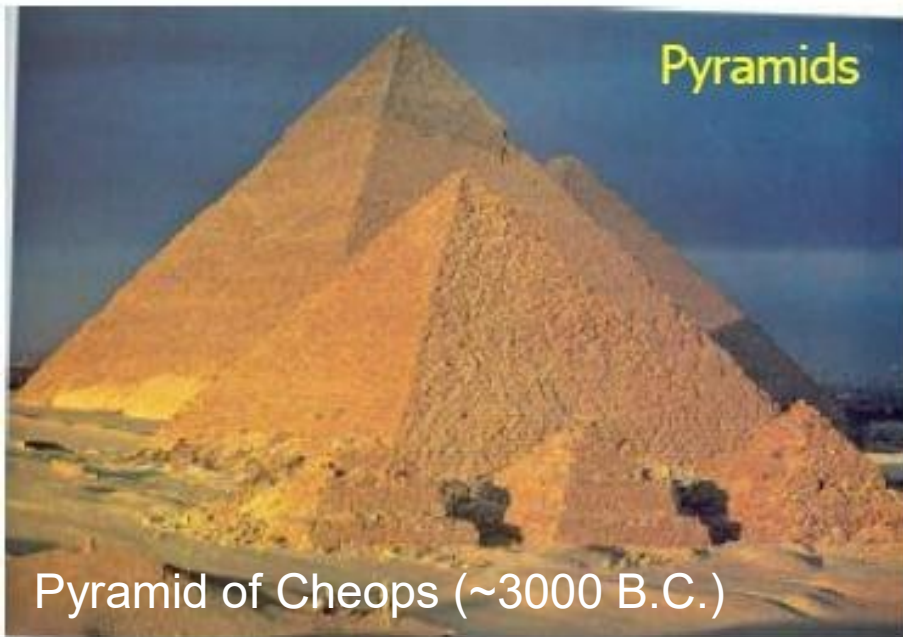
Nature of Concrete



Concrete cylinder full and sliced

Roles of Cement/Binder Paste

- **Coats** the aggregate particles
- **Fills the spaces** between aggregate particles
- **Binds** the aggregate particles
- Provides **strength & stiffness** to concrete
- Responsible for **permeability** and time-dependent **deformation** (shrinkage & creep)



**Egyptian concrete:
Gypsum and Lime**

**Roman Concrete - Binder
Pozzoulani sand, lime, and water**



Cement

(EN 197-1:2000)

- Cement is a **hydraulic binder**, i.e. a finely ground inorganic material which, when mixed with water, forms a paste which sets and hardens by means of hydration reactions and processes and which, after hardening, **retains its strength and stability even under water**

Raw Materials of Portland Cement

- Limestone, Chalk [$\text{CaCO}_3 = \text{CaO}$ (lime) + CO_2]

- Calcareous materials

- Clay, Shale – silica (SiO_2) & alumina (Al_2O_3)

- Argillaceous materials

- Iron Ore – Iron Oxide (Fe_2O_3)

- Flux to lower the clinkering temperature

Gypsum – Calcium Sulphate ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$)

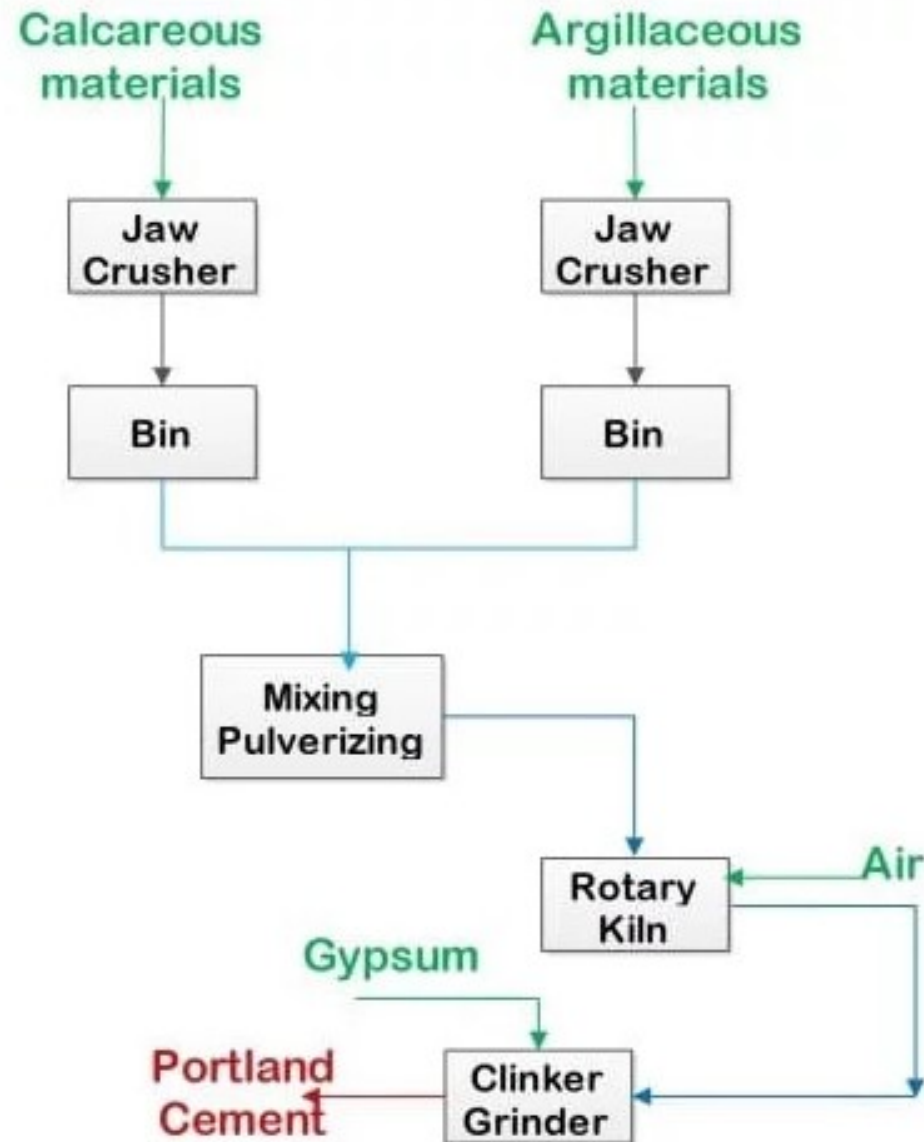
Control the rate of setting of cement

Note: The name 'Portland cement', given originally due to the resemblance of the color and quality of the hardened cement to Portland stone – a limestone quarried in Dorset, UK.

Manufacture of Portland Cement

- Cement, how it is made <http://youtu.be/n-Pr1KTVSXo>
- Collect the raw materials such as; limestone, clay, sea sand, shale, and etc.
- Grind the raw material into a very fine powder
- Mix them in predetermined proportions
- Burn them in a large rotary kiln at a temperature of about 1400°C when the material sinters and partially fuses into clinker
- The clinker is cooled and ground to a fine powder, with some gypsum added, and resulting product is the cement

Portland cement production



Clinker and Unhydrated Ordinary Portland Cement Particles



Major Chemical Compounds of Ordinary Portland Cement

Major Compounds

- **Tri-calcium silicate** [C_3S]
- $3CaO.SiO_2$
- **Di-calcium silicate** [C_2S]
- $2CaO.SiO_2$
- **Tri-calcium Aluminate** [C_3A]
- $3CaO.Al_2O_3$
- **Tetra-calcium Alumino-ferrite** [C_4AF]
- $4CaO.Al_2O_3.Fe_2O_3$

C: Calcium Oxide
(CaO)

S: Silicon Oxide
(SiO₂)

A : Aluminum
Oxide (Al₂O₃)

F: Iron Oxide
(Fe₂O₃)

Minor Chemical Compounds of Ordinary Portland Cement

- Alkalis (Soda Na_2O and Potash K_2O)

$$\text{Alkali Content} = \text{Na}_2\text{O} + 0.658 \text{ K}_2\text{O}$$

- Other Oxides: MgO ; Mn_2O_3 ; Free Lime

Minor refers to quantity only;

Significant effect on concrete durability

- Alkali aggregate reaction → Alkalis react with silica causing disintegration & expansion of concrete
- Soundness of cement
- Setting of cement

Composition Limits of Portland Cement

- Lime (C) 60 - 67%
- Silica (S) 17 - 25 %
- Alumina (A) 3 - 8 %
- Iron Oxide (F) 0.5 – 6.0 %
- Alkalis 0.3 - 1.2 %
- SO₃ (gypsum) 2.0 – 3.5%

	Usual Range by Weight (%)
C ₃ S	45 – 60
C ₂ S	15 – 30
C ₃ A	6 - 12
C ₄ AF	6 - 8

Bogue's Equations for Major compounds

$$C_3S = 4.07 C - 7.60 S - 6.72 A - 1.43 F - 2.85 SO_3$$

$$C_2S = 2.87 S - 0.75 C_3S$$

$$C_3A = 2.65 A - 1.69 F$$

$$C_4AF = 3.04 F$$

Example – Calculate the Bogue Composition of the Cements

	<i>Percentage in Cement No.</i>		
	(1)	(2)	(3)
Oxide			
CaO	66.0	63.0	66.0
SiO ₂	20.0	22.0	20.0
Al ₂ O ₃	7.0	7.7	5.5
Fe ₂ O ₃	3.0	3.3	4.5
Others	4.0	4.0	4.0
Compound			
C ₃ S			
C ₂ S			
C ₃ A			
C ₄ AF			

Influence of Change in Oxide Composition on Compound Composition

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Fe ₂ O ₃	3.0	3.3	4.5
Others	4.0	4.0	4.0
Compound			
C₃S	65	33	73
C₂S	8	38	2
C ₃ A	14	15	7
C ₄ AF	9	10	14

Minor changes in oxide composition changes C₃S:C₂S ratio significantly

Properties of Major Compounds

C₃ S: Hardens quickly; Heat evolved (500 J/g);

Early age strength; Light colour

C₂ S: Hardens slowly; Less heat evolved (250J/g) ;

Later age strength; Light colour

C₃ A: Reacts quickly; Heat evolved (850J/g);

Very early setting and strength gain;

Poor resistance to Sulphate; Light colour

C₄ AF: Reacts quickly; Heat evolved (400J/g);

Little strength; Dark colour

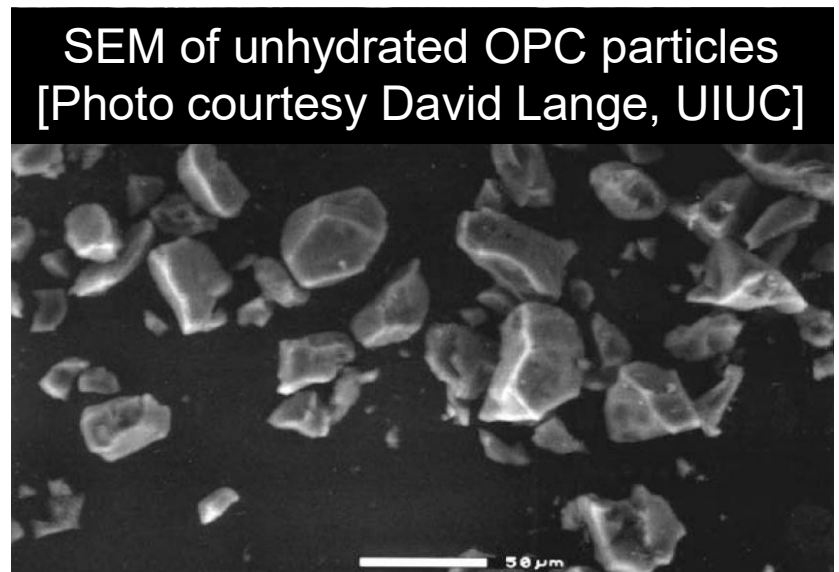
Physical Properties of Portland Cement

■ Specific gravity $G_{\text{cement}} \approx 3.15$

- Measured for cement particles without air voids
- Bulk unit weight (weight required to fill a container) is highly variable.
- Cement should not be measured by volume

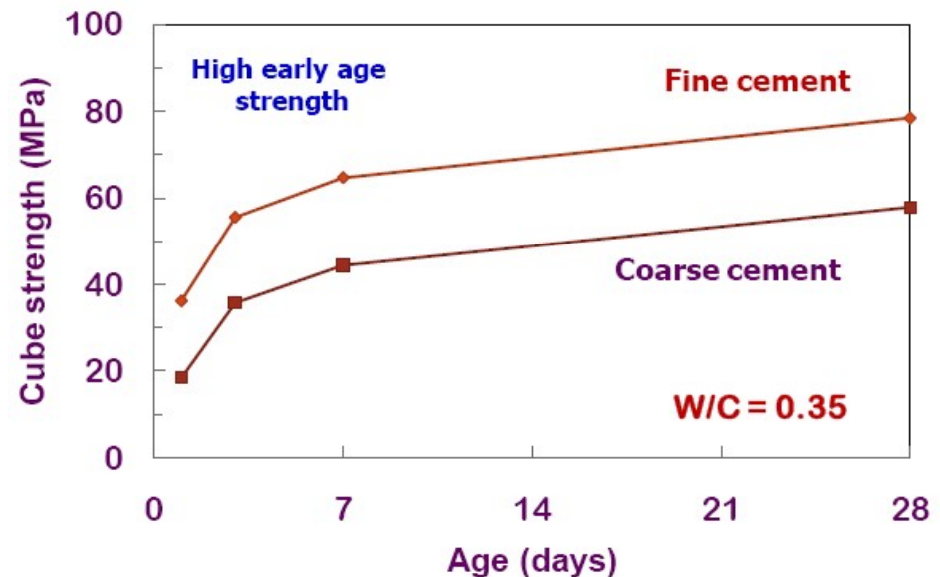
■ Particle shape: Angular

Particle Sizes: 0.1 to 90 μm



Fineness of Cement

- Smaller cement particles have **more surface area** to react with water
 - Fineness controls the rate of hydration (heat & strength gain)
 - Too fine is more expensive and can be harmful



Effect of Fineness on strength development in mortar

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